Hong Hu



PhD Applicant | Fall 2027 | Male | Age 24 (born January 2002)
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Research interests: AI for Science (AI4S), scientific agents, time-series forecasting

Education

Chongqing University of Science and Technology | Master's in Electronic Information

Sep 2024 – Jun 2027 | School of Computer Science and Engineering | Advisor: Guorong Chen

Research focus: Artificial Intelligence | GPA: 3.58/5.0

Thesis: Industrial Time-Series Forecasting with Causal Priors and Periodic Features

Peking University Shenzhen Graduate School | Joint Research Training

Apr 2026 – Present | School of Advanced Materials | Research advisor: Ruqiang Zou

Independently designed and developed DeepAssemble-Agent and HEO-Agent, including their computational analysis workflows, for single-crystal experiments and high-entropy oxide structure analysis. Collaborated with experimental researchers to incorporate observations into agent analysis and apply computational recommendations in subsequent experiments.

Chongqing University of Science and Technology | Bachelor's in Computer Science and Technology

Oct 2020 – Jun 2024

Research Experience

DeepAssemble-Agent | Agent for Materials Self-Assembly Experiments

- Independently built the complete system for well-plate image processing, state classification, experiment-history management and recipe recommendation, supporting batch analysis of 12 plates and 144 wells per round.
- Combined Hough circle detection and K-means grid localization to extract well images; implemented six-class whole-well classification with EfficientNet-B0. Tracked conditions and observation times by batch, well and bottle, distinguishing new and existing bottles and manual corrections.
- Converted image classifications, solvent ratios and cap settings into structured data for a large language model. Implemented parameter constraints and evidence checks to validate recommendations against experimental conditions and historical records.
- Collaborated with the experimental team on iterative planning. Recommendations for nearby solvent ratios and parallel-bottle validation have been used in real experiments. Further validation, analysis of new data and manuscript preparation are underway, with Nature as the intended submission venue.

HEO-Agent | Agent for High-Entropy Oxide Configurations and Spectroscopy

- Independently developed a computational system integrating configuration generation, MACE structure relaxation, FEFF spectroscopy simulations and posterior analysis, including batch scheduling, restart support, error checks and cross-stage data linkage.
- Combined configuration energies with experimental multi-edge XANES spectra. In the main batch workflow, estimated configuration posterior probabilities using spectral error and KL-divergence constraints; evaluated fits on held-out contiguous energy intervals.
- Completed large-scale result compilation and posterior analysis for two series, totaling 2,144,454 configuration-probability records. Produced probability distributions, configuration sets covering 95% cumulative posterior probability and spectral comparison reports.
- Collaborated with experimental researchers to incorporate measured spectra into the analysis, supporting comparisons of candidate configurations and subsequent structural interpretation.

P2D-Former | Time-Series Forecasting with Periodic Structure and Exogenous Variables

Independently designed and implemented the model and conducted experiments; published as first author in Scientific Reports.

- Encoded the target series as patches, extracted dominant periods using the fast Fourier transform, and reshaped sequences into 2D representations. Used multi-scale convolutions to model intra-period and inter-period dependencies.
- Designed Intersect-Attention to query exogenous variables using a global target-series representation and fuse the resulting information back into local sequence representations.
- Conducted long- and short-term forecasting, imputation and ablation experiments. On Weather, average MSE across four forecast horizons was 0.239 versus 0.258 for iTransformer, a relative reduction of approximately 7.4%.

Industrial Gas Monitoring | Multi-Sensor Time-Series Modeling and False-Alarm Identification

Master's research | First-author publication and thesis work

- Integrated meteorological and gas-sensor data; used Pearson correlation, LOWESS and quantile regression to examine relationships between environmental variables and gas readings. Developed alarm-identification rules incorporating wind speed and direction; published at IEEE IICSPI 2025.
- Processed 183 days of data from six gas-monitoring points and one weather station: 2,773,862 deduplicated observations, 24,158 alarm events and a unified 4,392-hour time grid.
- Used chronological train/validation/test splits and fitted scaling parameters on training data only. Retained missingness indicators and observation time gaps to avoid future-data imputation. Completed preprocessing and forecast-window checks; industrial forecasting experiments are ongoing.

Selected Publications

1. Hu Hong, Guorong Chen, Wang Jian, et al. P2D-former for time-series forecasting with period-aware 2D representations and exogenous covariates. Scientific Reports, 16, 17218, 2026.

DOI: 10.1038/s41598-026-45499-w

2. Hong Hu, Jingcheng Liu, Bao Zhang, et al. A Method for False Alarm Identification in Industrial Gas Detection Systems through the Integration of Meteorological and Multi-Sensor Data. IEEE IICSPI 2025, 208–215.

DOI: 10.1109/IICSPI66775.2025.11437860

Projects and Awards

- Project lead for municipal and university postgraduate research innovation projects: "Combustible Gas Leakage Situation Awareness Using 4D Maps" (CYS25836 and YKJCX2421508); completed in June 2026.
- Led the preparation of team project applications, technical proposals and completion reports. The five related projects had a cumulative team contract value of approximately RMB 5 million.
- Team leader, 6th Global Campus Artificial Intelligence Algorithm Elite Competition (2024): national third prize and provincial first prize.
- First-Class Postgraduate Scholarship.

Technical Skills and English

Python, PyTorch; time-series Transformers, image classification, LLM tool calling and structured outputs; MACE/FEFF workflow integration, batch job management and posterior analysis.

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